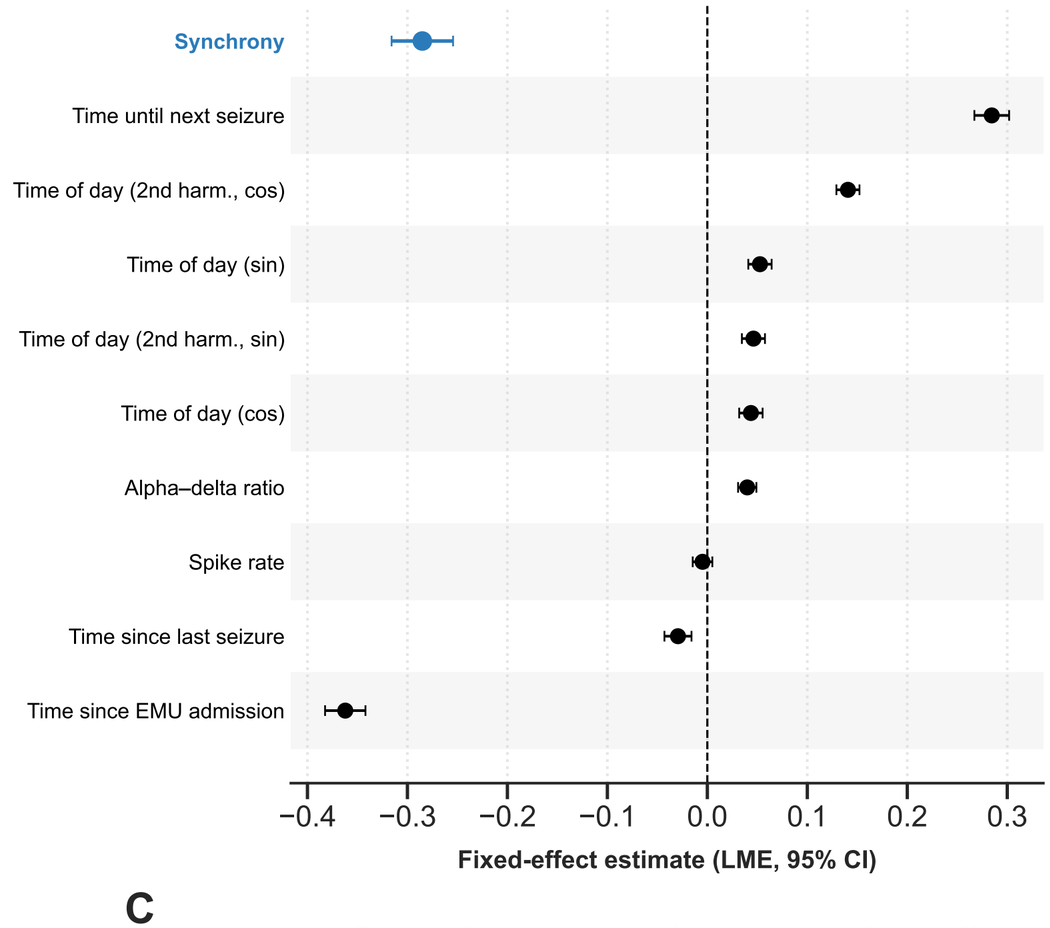
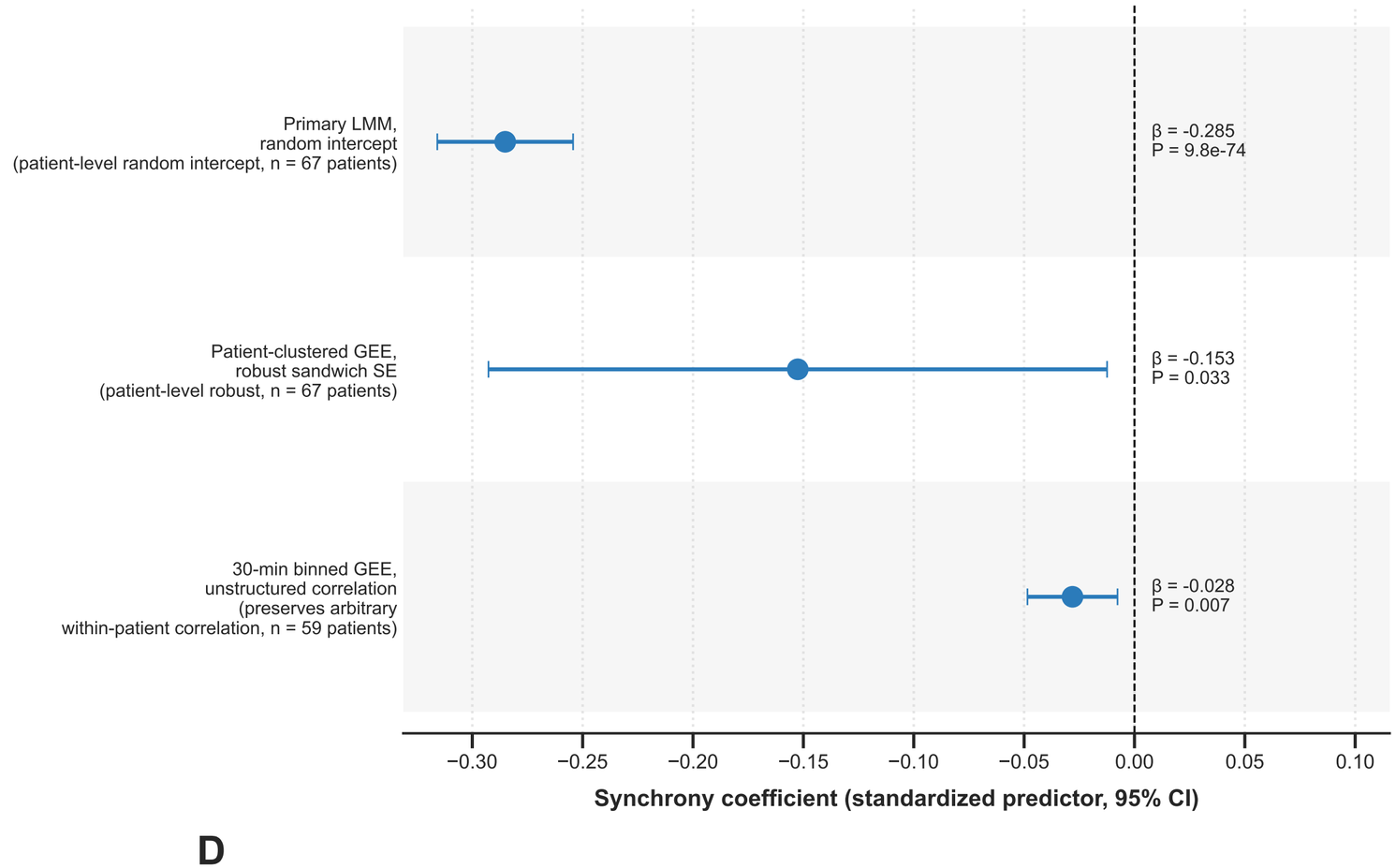
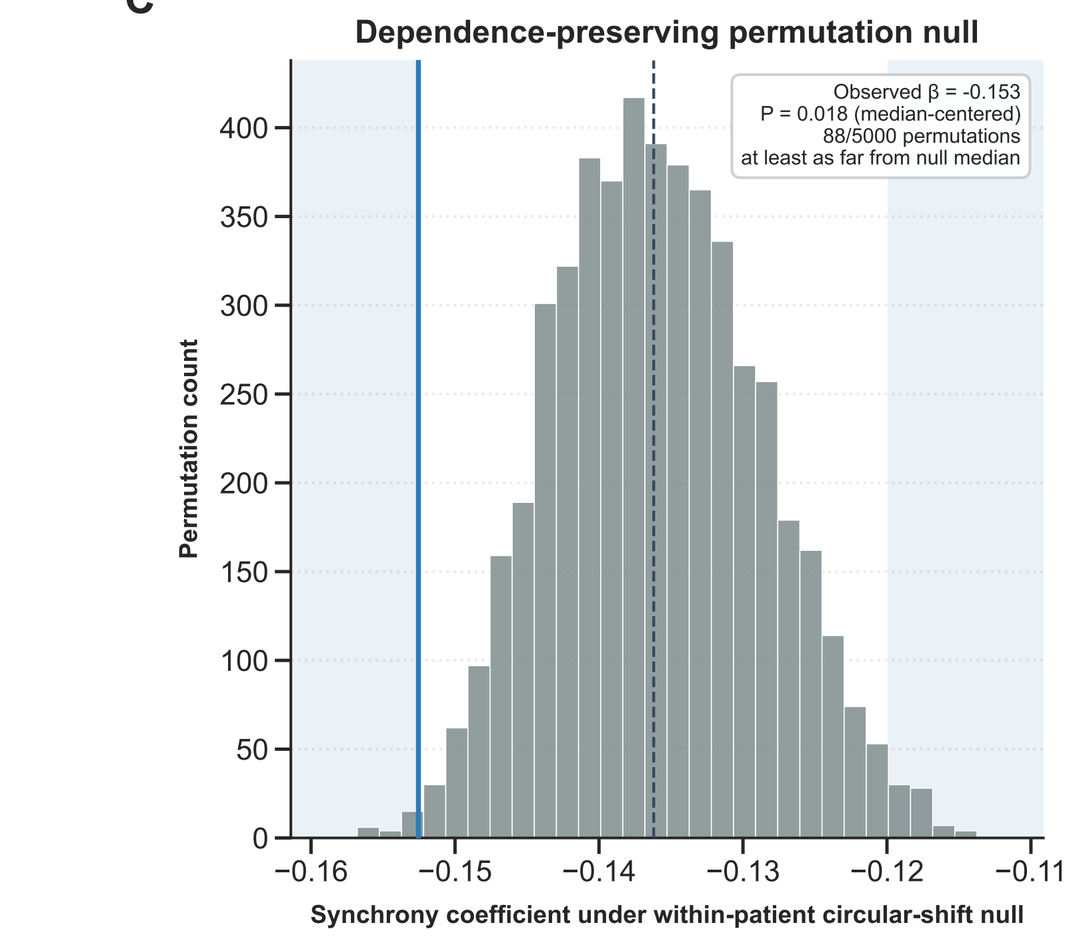
A


Linear Mixed-Effects Model
Synchrony
Time until next seizure
Time of day (2nd harm., cos)
Time of day (sin)
Time of day (2nd harm., sin)
Time of day (cos)
Alpha–delta ratio
Spike rate
Time since last seizure
Time since EMU admission
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Fixed-effect estimate (LME, 95% CI)

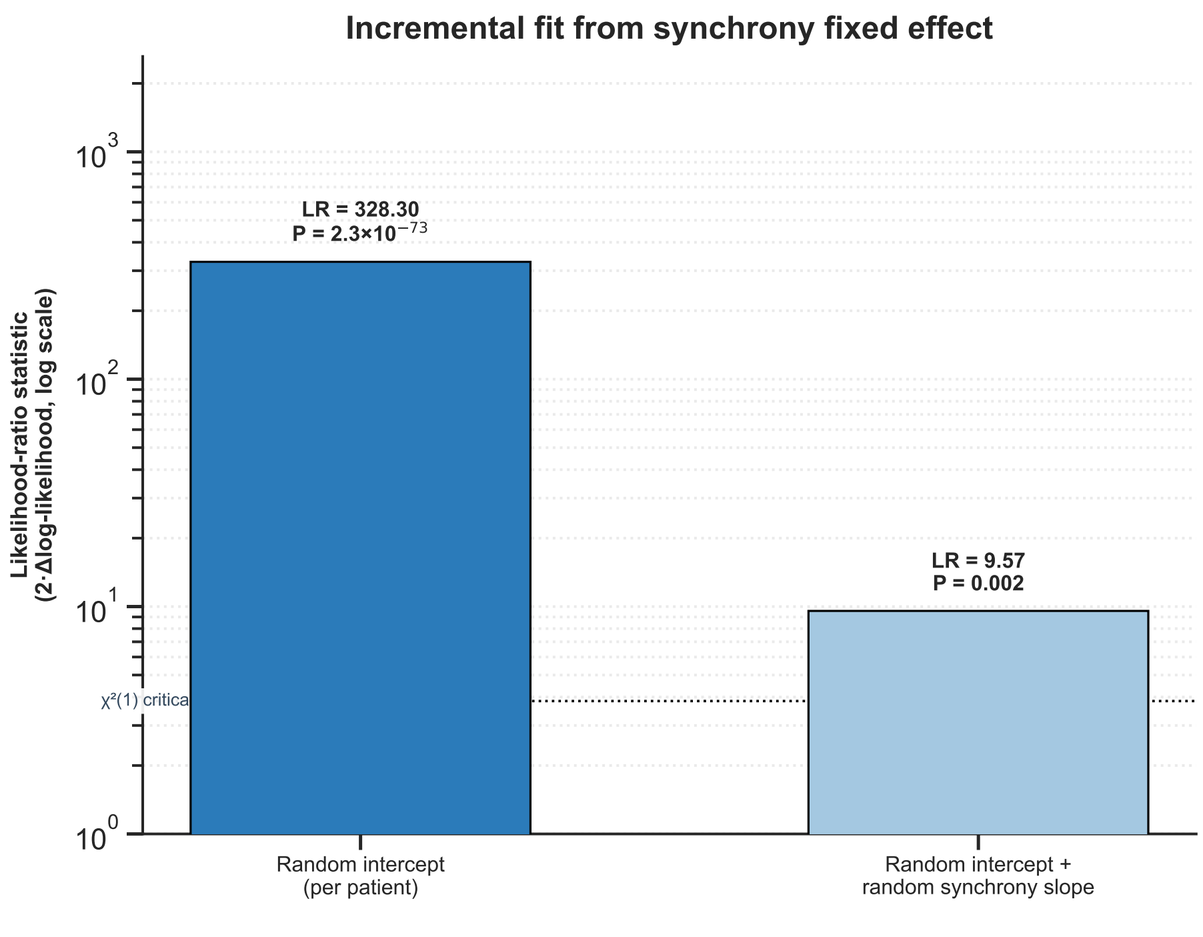
B


Synchrony coefficient across longitudinal models
Primary LMM,
random intercept
(patient-level random intercept, n = 67 patients)
β = -0.285
P = 9.8e-74
Patient-clustered GEE,
robust sandwich SE
(patient-level robust, n = 67 patients)
β = -0.153
P = 0.033
30-min binned GEE,
unstructured correlation
(preserves arbitrary
within-patient correlation, n = 59 patients)
β = -0.028
P = 0.007
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
Synchrony coefficient (standardized predictor, 95% CI)

C


Dependence-preserving permutation null
Observed β = -0.153
P = 0.018 (median-centered)
88/5000 permutations
at least as far from null median
Permutation count
0
50
100
150
200
250
300
350
400
−0.16
−0.15
−0.14
−0.13
−0.12
−0.11
Synchrony coefficient under within-patient circular-shift null

D


Incremental fit from synchrony fixed effect
Likelihood-ratio statistic
(2·Δlog-likelihood, log scale)
10^3
10^2
10^1
10^0
LR = 328.30
P = 2.3×10^−73
LR = 9.57
P = 0.002
χ²(1) critica
Random intercept
(per patient)
Random intercept +
random synchrony slope